



Shishu-Sneh

Baby's First Year Guide

Product Requirements Document

Final Version

Prepared By:	Pallavi Asnotikar
USN:	4SN22CB035
College:	Shrinivas Institute of Technology
Department:	CS & Business Studies
Domain:	Android App Dev — Healthcare
Platform:	Android

1 Executive Summary

Shishu-Sneh is an **offline-first maternal and infant healthcare Android application** designed to support new mothers during the baby's first year of life. Acting as a "*digital elder*", the app helps mothers monitor baby growth, vaccination schedules, developmental milestones, feeding habits, sleep tracking, diaper tracking, medication reminders, and AI-assisted healthcare guidance.

Specially crafted for **rural and semi-urban mothers** who may lack continuous hospital guidance after childbirth, Shishu-Sneh combines modern Android architecture with healthcare usability principles to create a reliable, scalable, and emotionally supportive companion.

- Child healthcare awareness
- Vaccination tracking
- Growth monitoring
- Local accessibility
- Offline healthcare support
- AI-assisted parenting guidance
- Mother-friendly UX
- Secure data management

2 Problem Statement

Many new mothers in rural and semi-urban regions face significant healthcare gaps after leaving the hospital. Without a reliable support system, parents struggle to:

- Track vaccination schedules and receive timely reminders
- Monitor baby growth trends and identify abnormalities early
- Record and manage daily feeding routines
- Understand and verify developmental milestones
- Store and access baby healthcare records safely
- Receive healthcare guidance in local languages

These gaps contribute to missed vaccinations, improper nutrition, delayed milestone identification, and increased infant health risks — particularly in communities with limited access to continuous professional healthcare.

3 Vision & Objectives

"To create a trustworthy digital healthcare companion that empowers mothers with accessible, reliable, and emotionally supportive infant care guidance during a baby's most critical developmental years."

Primary Objectives

- Help mothers track baby growth and development
- Provide automated, reliable vaccination reminders
- Encourage proper nutrition and feeding practices
- Support healthcare accessibility in local languages
- Maintain secure, offline healthcare records
- Reduce missed vaccination schedules

Secondary Objectives

- Create a calm and mother-friendly healthcare UX

- Integrate AI-assisted healthcare guidance via Gemini API
- Provide emotional reassurance through healthcare insights
- Build a scalable healthcare monitoring platform

4 Target Users

Primary Users	New mothers, rural mothers, semi-urban families, parents with infants aged 0–12 months
Secondary Users	Healthcare workers, Anganwadi workers, family caregivers, pediatric support staff

5 Core Features

5.1 Baby Profile Management

Create and manage baby profiles with name, date of birth, gender, birth weight, mother details, location, and preferred language. Profile data drives vaccination schedules, growth analytics, milestone recommendations, and personalised guidance.

5.2 Growth Tracking System

Log monthly weight and height records, view growth trend visualisations (MPAndroidChart), and receive healthy-growth indicators to help identify abnormal patterns early.

5.3 Vaccination Reminder System

Automatically generate immunisation calendars from the baby's date of birth. Persistent notifications via WorkManager cover vaccines including Polio, BCG, DPT, Hepatitis B, and Measles.

5.4 Feeding & Nutrition Guide

Daily nutrition tips, feeding logs, AI-assisted feeding guidance, exclusive breastfeeding awareness content, and optional voice guidance support for low-literacy users.

5.5 Developmental Milestone Tracking

Checklist-based yes/no milestone verification with progress percentages. Covers milestones such as smiling, head control, crawling, sitting, and early speech.

5.6 Sleep Tracker

Log sleep entries and monitor sleep history to identify patterns and encourage healthy sleep habits.

5.7 Diaper Tracker

Maintain daily diaper-change logs and history records for hygiene awareness and routine management.

5.8 Medication Reminder System

Log medications, schedule dosage reminders, and track active medicines to support treatment consistency.

5.9 Medical Vault

Securely store prescriptions and health records for easy, organised access to the baby's complete healthcare history.

5.10 Ask Sneha AI Assistant

Gemini API-powered conversational assistant providing simplified healthcare Q&A; on feeding, vaccination, and parenting. Safety-first design: does not replace professional medical advice.

5.11 Voice Guidance System

Text-to-Speech narration of healthcare tips and reminders, improving accessibility for low-literacy and rural users.

5.12 Multi-Language Support

Full support for English, Kannada, and Hindi to maximise inclusivity across the target user base.

6 System Architecture

The application follows **Clean Architecture** principles with three distinct layers ensuring separation of concerns, testability, and scalability.

Layer	Components
Presentation Layer	Jetpack Compose UI • Navigation Components • ViewModels • State Management
Domain Layer	Business Logic • Use Cases • Healthcare Processing Logic
Data Layer	Room Database • DAO Classes • Repositories • Encrypted Local Storage

7 Technology Stack

Component	Technology
Programming Language	Kotlin
UI Framework	Jetpack Compose
Architecture Pattern	Clean Architecture + MVVM
Database	Room DB
Dependency Injection	Hilt
Notifications	WorkManager
Charts	MPAndroidChart
AI Integration	Gemini API
Voice Guidance	Android TextToSpeech
Storage Encryption	SQLCIPHER
Navigation	Navigation Compose

8 Database Design

The Room database is **encrypted via SQLCIPHER** and designed with an offline-first strategy. All entities maintain foreign key relationships for data integrity.

- BabyEntity
 - GrowthEntity
 - VaccineEntity
 - MilestoneEntity
 - UserProfileEntity
- FeedingEntity
 - SleepEntity
 - DiaperEntity
 - MedicationEntity

9 UI / UX Design Principles

Shishu-Sneh adopts a **soft, calm, and emotionally supportive design language** tailored for first-time mothers, including those in rural settings.

Colour Palette	Soft pastels — blush pinks, warm creams, gentle greens
Typography	Calm, rounded fonts with large, readable text
Cards & Layout	Rounded cards, ample white space, minimal visual clutter
Navigation	Bottom navigation for single-thumb, mother-friendly access
Accessibility	High-contrast text, TTS support, large tap targets
Emotional Design	Reassuring copy, encouraging progress indicators

10 Security & Privacy

- SQLCipher-encrypted Room database for all baby health data
- Offline-first architecture minimises cloud data exposure
- All personal data stored locally on-device
- No third-party data sharing or advertising integration
- Secure data persistence across app sessions

11 Future Enhancements

- WHO growth percentile overlays
- OCR vaccination card scanning
- Cloud backup support
- Pediatric consultation integration
- Improved AI healthcare assistant
- Enhanced multilingual AI responses
- Advanced voice interaction
- Advanced healthcare analytics

12 Social Impact

Shishu-Sneh directly addresses healthcare inequality by providing accessible digital healthcare assistance to underserved communities. The application contributes toward:

- Child healthcare awareness and vaccination consistency
- Infant nutrition support and breastfeeding guidance
- Rural healthcare accessibility and maternal support
- Healthcare literacy improvement in low-resource settings
- Child survival improvement through timely interventions

13 Success Criteria

Growth Charts	Growth trends display correctly with MPAndroidChart visualisations
Vaccination Reminders	Reminders fire automatically and persist via WorkManager
Healthcare Records	All records stored securely and retrievable offline
UI Quality	Mother-friendly, readable, and accessible interface confirmed
Voice Support	TTS narration improves accessibility for low-literacy users
AI Guidance	Gemini API assistant provides relevant parenting responses
Offline Functionality	Full feature set available without internet connectivity

14 References

- [1] Android Developer Documentation
- [2] Jetpack Compose Documentation
- [3] Room Database Documentation

- [4] WorkManager Documentation
- [5] Google Gemini API Documentation
- [6] MPAndroidChart Documentation
- [7] WHO Child Growth Standards
- [8] UNICEF Child Healthcare Guidelines